

Project Approach (Roadmap)

Title: Comparative Analysis of Explainable Machine Learning Models for Business Analytics

Domain

Business Analytics

Dataset

I recommend using **three benchmark datasets together**:

1. UCI Bank Marketing Dataset (UCI Machine Learning Repository)

<https://archive.ics.uci.edu/dataset/222/bank%2Bmarketing>

2. Telco Customer Churn Dataset for Customer Analytics

https://www.kaggle.com/datasets/blaschar/telco-customer-churn/data?utm_source=

3. UCI Online Retail Dataset for E-commerce and customer transaction analytics

<https://archive.ics.uci.edu/dataset/352/online-retail>

All the 3 are famous Valid Datasets. All are to be considered in the Project.

Abstract

Business Analytics plays a vital role in helping organizations make data-driven decisions by extracting meaningful insights from large volumes of data. However, many high-performing machine learning models operate as "black boxes," making it difficult for decision-makers to understand how predictions are generated. This project aims to perform a comparative analysis of explainable machine learning models to evaluate both predictive performance and interpretability in business analytics applications. The study utilizes three benchmark datasets: the UCI Bank Marketing Dataset, the Telco Customer Churn Dataset, and the UCI Online Retail Dataset, representing different real-world business scenarios.

The proposed methodology begins with Exploratory Data Analysis (EDA), including dataset overview, data quality assessment, descriptive statistics, missing value handling, duplicate removal, outlier detection, and univariate, bivariate, and multivariate analysis. The data is then preprocessed through cleaning, encoding, normalization, feature engineering, feature selection, and train-test splitting to prepare it for machine learning model development.

Multiple classification algorithms, including Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost, are trained and evaluated. Their performance is measured using standard evaluation metrics such as Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, Confusion Matrix, and Classification Report. To improve model transparency and trustworthiness, Explainable Artificial Intelligence (XAI) techniques such as SHAP and LIME are applied to provide both global and local explanations of model predictions and identify the most influential features affecting business decisions.

The project also includes visualization techniques such as heatmaps, boxplots, correlation matrices, feature importance graphs, SHAP visualizations, LIME visualizations, and an interactive dashboard for effective presentation of analytical results. Finally, a comparative analysis is conducted to examine the strengths, weaknesses, predictive performance, and explainability of each model, enabling the selection of the most suitable model for business analytics applications. The proposed framework demonstrates how explainable machine learning can enhance both prediction accuracy and decision transparency, making AI systems more reliable and practical for real-world business environments.

Data Analytics

- EDA

➔ EDA Flow Diagram:

Raw Dataset



Dataset Overview



Data Quality Assessment



Descriptive Statistics



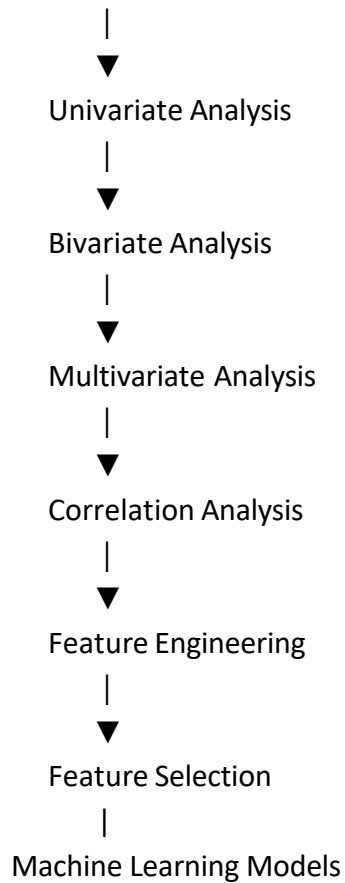
Missing Values



Duplicate Records



Outlier Detection



Preprocessing

- Data Cleaning
- Missing Value Handling
- Duplicate Removal
- Data Encoding
- Feature Scaling / Normalization
- Train-Test Split

ML Models

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- CatBoost
- Model Comparison

Explainability

- SHAP Analysis
- LIME Analysis

Visualization

- Heatmaps
- Boxplots
- Correlation Matrix
- Feature Importance Graphs
- SHAP Visualizations
- LIME Visualizations
- Interactive Dashboard

Comparative Analysis

Compare

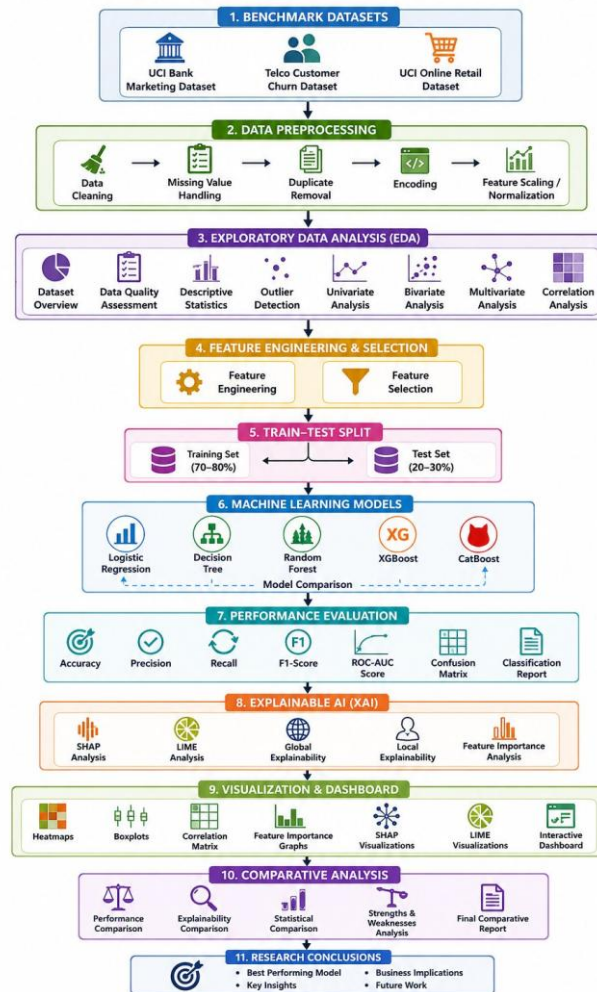
- Performance Comparison of Models
- Explainability Comparison
- Statistical Comparison
- Strengths and Weaknesses Analysis
- Final Comparative Report

Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score
- Confusion Matrix
- Classification Report

ARCHITECTURE FLOW DIAGRAM

Comparative Analysis of Explainable Machine Learning Models for Business Analytics



Recommended Research Pipeline:

Benchmark Dataset



Data Cleaning



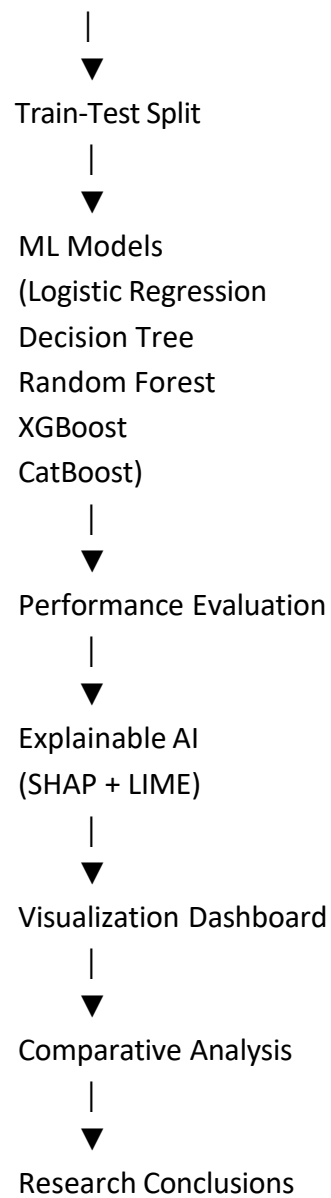
EDA



Feature Engineering



Feature Selection



Batch-2

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2420030508

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